MUHAMMAD ASAD FEROZI

Applicant, MS in Computer Science (Artificial Intelligence) · Fall 2027

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RESEARCH INTERESTS

Applied machine learning and large language models; retrieval-augmented generation and grounded question answering; trustworthy and secure AI; computer vision for assistive technology; real-time and distributed systems for AI applications.

I am a software engineer at INTECH, building EmpowerGPT, an enterprise generative-AI platform: assistant builders, hybrid document search and multi-tenant sign-in. I hold a BS in Information Technology (CGPA 3.78/4.00) and am seeking a research-based master's to study how to make LLM systems grounded, measurable and safe to connect to real tools.

EDUCATION

Bachelor of Science in Information Technology (4 years, 134 credit hours) 2021 – 2025

Punjab University College of Information Technology (PUCIT), University of the Punjab, Lahore **CGPA 3.78 / 4.00 · 85.40%**

- Final-semester GPA 3.94; first division. English was the medium of instruction.
- Capstone project: *Parental Eye: Child Tracking System*. Grades: Capstone I A (85), Capstone II A (90).

Design & Analysis of Algorithms	A · 90	Artificial Intelligence	A · 85	Computer Organization & Design	A · 93
Discrete Structures	A · 91	Linear Algebra	A · 85	Computer Networks	A · 90
Data Structures & Algorithms	A · 85	Calculus & Analytical Geometry	A · 86	Cyber Security	A · 85
Programming Fundamentals	A · 94	Probability & Statistics	A– · 80	Information Security	A– · 82
Software Quality Assurance	A · 89	Machine Learning	B · 73	Operating Systems	A– · 81

Intermediate (FSc Pre-Engineering), Board of Intermediate & Secondary Education, Multan 2021

Punjab College for Boys, Burewala **1071 / 1100 (97.4%) · A+**

Matriculation (Science), Board of Intermediate & Secondary Education, Multan 2019

Govt. Higher Secondary School Sheikh Fazal, Burewala **1049 / 1100 (95.4%) · A+**

PROFESSIONAL EXPERIENCE

Software Engineer, INTECH Automation and Intelligence, Lahore Jun 2026 – Present

- Led the AI Assistants revamp for **EmpowerGPT**: REST APIs for creating and sharing custom assistants, enterprise-knowledge and image-generation settings, and a redesigned assistant builder and knowledge-upload flow.
- Built Advanced Search and Explorer end to end (FastAPI, React): hybrid search across enterprise documents with suggestions, recent-activity tracking and workspace navigation, backed by indexing on multi-tenant PostgreSQL tables.

Associate Software Engineer, INTECH Automation and Intelligence, Lahore Jun 2025 – Jun 2026

- Led the Data Library revamp across three versions: file and folder browsing with permissions, uploads for CSV, audio and video with size limits and cancelling, and translations.
- Designed multi-tenant sign-in on Keycloak: domain-based tenant discovery, credential-reset rules for new tenants and Azure AD language sync for enterprise SSO.
- Implemented rate limiting for AI image generation across the chat and core backend services.
- Drove a configuration refactor across about 10 microservices and the frontend for multi-environment Kubernetes deployments, and contributed to the shared Python framework used by every backend service.
- Authored 145 pull requests (135 merged) on the platform since October 2025.

Full-stack Developer, Webox Systems, Lahore Jul 2024 – Jul 2025

- Developed **Euthyna**, a sustainability analytics platform that automates ESG gap analysis and compliance checks (PostgreSQL, Express, React, Node.js).
- Code-splitting, database indexing and API optimisation cut page load times by 30%; designed PostgreSQL schemas across 15+ tables, improving query efficiency by 30%.

PERN Stack Fellow, Webox Systems, Lahore · **Frontend Fellow**, Bytewise Limited (remote) 2023 – 2024

- Two fellowships (Mar–Jul 2023, Feb–May 2024) building full-stack and React applications in Agile teams.

RESEARCH AND ACADEMIC PROJECTS

Parental Eye: Real-Time GPS Tracking with Geofencing · Final-year project, team of four Oct 2024 – Jun 2025

React · Leaflet · Node.js · MQTT · Socket.IO · PostgreSQL/PostGIS · Turf.js · parental-eye-react.vercel.app

- Devices publish locations over MQTT; the server checks each point against circle, polygon and route geofences with Turf.js, stores it in PostGIS, and pushes alerts over Socket.IO to users in five roles.
- Added alert de-duplication and device online/offline detection; built map-based geofence drawing and role dashboards in React.

AI Lawyer: Retrieval-Augmented Legal Assistant · Personal project Jun 2024 – Oct 2024

React · Express · LangChain · OpenAI embeddings · Pinecone · PostgreSQL

- Legal PDFs (Pakistan Penal Code, a criminal-investigation handbook) are chunked, embedded and indexed in Pinecone; each question retrieves the closest passages and a LangChain QA chain answers from them, with chat history stored per user.

Be My Eye: Obstacle Detection for the Visually Impaired Jan 2024 – Apr 2024

Python · Computer vision · Roboflow · Google Colab

- Trained a real-time obstacle-detection model reaching 90% accuracy, 92% precision and 91% recall.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Java, C++, PHP
AI and Machine Learning	Retrieval-augmented generation, LangChain, OpenAI API, vector databases (Pinecone), hybrid search, computer vision
Back End	FastAPI, Node.js, Express, REST, WebSockets, MQTT, rate limiting, microservices
Data	PostgreSQL (partitioning, indexing), PostGIS, Redis, MongoDB
Front End	React, Redux Toolkit, Tailwind CSS, AG Grid, Leaflet
Platform	Kubernetes, Docker, Keycloak, Azure AD SSO, GitHub Actions, Nginx, Linux

CERTIFICATION

Supervised Machine Learning: Regression and Classification, Stanford University and DeepLearning.AI (Coursera), January 2025

LANGUAGES

English: professional working proficiency; all four years of the BS were taught in English (Medium of Instruction certificate available). **Urdu:** native.

REFERENCES

Available on request.